# Lukas Folkman, PhD

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## Research Interests

**Applied machine learning**

- Deep learning for single-cell transcriptomics and molecular property prediction
- Personalised drug recommendation systems
- Prediction of disease-causing and functionally-relevant genomic variants

**Data science for the life sciences**

- Integrative statistical analysis of high-dimensional molecular data

## Education

2011–2015 **PhD Candidate**[1,2]
[1]Institute for Integrated and Intelligent Systems, Griffith University, Australia
[2]NICTA – Australia's ICT Research Centre of Excellence, Australia
Advisers: Prof Bela Stantic[1], Prof Yaoqi Zhou[1], Prof Abdul Sattar[1,2]
Project: Predicting stability and functional changes induced by protein mutations with a machine learning approach

2007–2010 **Student of Informatics (Master's Degree)**
Faculty of Informatics, Masaryk University, Czech Republic
Awards: Dean's awards for excellent study results and outstanding thesis

2007 **Visiting Student**
School of Computer Science, University College Dublin, Ireland
Awards: Erasmus Student Exchange Scholarship

2004–2007 **Student of Applied Informatics (Bachelor's Degree)**
Faculty of Informatics, Masaryk University, Czech Republic
Awards: Average grade ranked within the best 2% of 2006 and 2007 graduates

## Professional Experience

2018–now **Postdoctoral Researcher**
CeMM Research Center for Molecular Medicine, Austrian Academy of Sciences, Austria
Adviser: Prof Christoph Bock (https://www.bocklab.org)
Projects: Deep learning for single-cell transcriptomics
Deep learning for molecular property prediction
Statistical and bioinformatics analysis of high-dimensional molecular data

2016–2018 **Postdoctoral Researcher**
Department of Biosystems Science and Engineering, ETH Zurich, Switzerland
Adviser: Prof Karsten Borgwardt (https://bsse.ethz.ch/mlcb)
Project: Machine learning for personalised drug recommendation systems

2015–2016 **Visiting Researcher**
Institute for Integrated and Intelligent Systems, Griffith University, Australia
Adviser: Prof Yaoqi Zhou (https://sparks-lab.org)
Project: Machine learning for prediction of disease-causing genomic variants

2008–2010 **Software Developer and Consultant**
ARTIN, Czech Republic

## Funding

2019–2021 Marie Skłodowska-Curie Actions Individual Fellowship (186,000 EUR)
Success rate of 12.4%; competitive fellowship sponsored by the European Commission
Project: Deep learning for analysis of single-cell transcriptomics

2011–2014 NICTA Project Award Scholarship (28,000 AUD)

2011–2014 Griffith University Postgraduate Research Scholarship (74,000 AUD+tuition fees waiver)

Dec 2010 NICTA Summer Research Scholarship (5,000 AUD)

## Awards

Sep 2017 Online Poster Award (1st place) at [BC]$^2$ Basel Computational Biology Conference 2017 for "Kernelized rank learning for personalized drug recommendation", Basel, Switzerland

Jul 2017 F1000Research Poster Award at the international conference ISMB/ECCB 2017 for "Kernelized rank learning for personalized drug recommendation", Prague, Czech Rep.

Jun 2014 Winner of Three-Minutes Thesis Competition, School of ICT, Griffith University

Apr 2014 Gold Coast Association of Postgraduates Conference Funding Award, Griffith University

Jan 2013 Griffith University Conference Travel Grant, Griffith University

Dec 2010 2nd best paper in the workshop Keymaker 2010, Brno, Czech Rep.

May 2010 2nd place in Czech and Slovak university research competition, Ostrava, Czech Rep.

Mar 2010 Dean's award for outstanding thesis, Faculty of Informatics, Masaryk University

Mar 2010 Dean's award for excellent study results, Faculty of Informatics, Masaryk University

## First-Author Publications

**# These authors contributed equally**

He[#] X, **Folkman[#] L**, Borgwardt K: "Kernelized rank learning for personalized drug recommendation".
*Bioinformatics* (2018), 34(16), 2808–2816. 🕮

Livingstone[#] M, **Folkman[#] L**, Yang Y, Zhang P, Mort M, Cooper DN, Liu Y, Stantic B, Zhou Y: "Investigating DNA-, RNA-, and protein-based features as a means to discriminate pathogenic synonymous variants".
*Human Mutation* (2017), 38(10), 1336–1347. 🕮

**Folkman L**, Stantic B, Sattar A, Zhou Y: "EASE-MM: sequence-based prediction of mutation-induced stability changes with feature-based multiple models".
*Journal of Molecular Biology* (2016), 428(6), 1394–1405. 🕮

**Folkman L**, Yang Y, Li Z, Stantic B, Sattar B, Mort M, Cooper DN, Liu Y, Zhou Y: "DDIG-in: detecting disease-causing genetic variations due to frameshifting indels and nonsense mutations by sequence and structural properties at nucleotide and protein levels".
*Bioinformatics* (2015) 31(10), 1599–1606. 🕮

**Folkman L**, Stantic B, Sattar A: "Feature-based multiple models improve classification of mutation-induced stability changes".
*BMC Genomics* (2014) 15(Suppl 4), S6. 🕮

**Folkman L**, Stantic B, Sattar A: "Towards sequence-based prediction of mutation-induced stability changes in unseen non-homologous proteins".
*BMC Genomics* (2014) 15(Suppl 1), S4. 🕮

**Folkman L**, Stantic B, Sattar, A: "Sequence-only evolutionary and predicted structural features for the prediction of stability changes in protein mutants".
*BMC Bioinformatics* (2013) 14(Suppl 2), S6. 🕮

## Co-Author Publications

de Bree LCJ, Mourits VP, Koeken VACM, Moorlag SJCFM, Janssen R, **Folkman L**, …, Netea MG: "Circadian rhythm influences induction of trained immunity by BCG vaccination".
*The Journal of Clinical Investigation* (2020), 130(10), 5603–5617. 🕮

Pejaver V, Babbi G, Casadio R, **Folkman L**, Katsonis P, Kundu K, …, Bromberg Y: "Assessment of methods for predicting the effects of PTEN and TPMT protein variants".
*Human Mutation* (2019), 40(9), 1495–1506. 🕮

Savojardo C, Petrosino M, Babbi G, Bovo S, Corbi-Verge C, Casadio R, Fariselli P, **Folkman L**, …, Capriotti E: "Evaluating the predictions of the protein stability change upon single amino acid substitutions for the FXN CAGI5 challenge".
*Human Mutation* (2019), 40(9), 1392–1399. 🕮

Clark WT, Kasak L, Bakolitsa C, Hu Z, Andreoletti G, Babbi G, Bromberg Y, Casadio R, Dunbrack R, **Folkman L**, …, LeBowitz JH: "Assessment of predicted enzymatic activity of alpha-N-acetylglucosaminidase (NAGLU) variants of unknown significance for CAGI 2016".
*Human Mutation* (2019) 40(9), 1519–1529. 🕮

Stetsko A, **Folkman L**, Matyas V: "Neighbor-based intrusion detection in wireless sensor networks".
*in* 'Wireless and Mobile Communications (ICWMC)', pp. 420–425, IEEE, 2010. 🕮

## Oral Presentations

2014 Asia-Pacific Bioinformatics Conference (APBC), Vancouver, Canada
"Towards sequence-based prediction of mutation-induced stability changes in unseen non-homologous proteins"

2013 Conference on Intelligent Systems for Molecular Biology & European Conference on Computational Biology (ISMB/ECCB), SNP Special Interest Group, Berlin, Germany
"Feature-based multiple models improve classification of mutation-induced stability changes"

2013 Asia-Pacific Bioinformatics Conference (APBC), Vancouver, Canada
"Sequence-only evolutionary and predicted structural features for the prediction of stability changes in protein mutants"

2011 Advances of Distributed Computing and Networking Symposium, Melbourne, Australia
"Generic parallel genetic algorithm framework for protein optimisation"

## Poster Presentations

2018 German Conference on Bioinformatics, Vienna, Austria
"Kernelized rank learning for personalized drug recommendation"

2017 $[BC]^2$ Basel Computational Biology Conference, Basel, Switzerland
"Kernelized rank learning for personalized drug recommendation"

2017 International Conference on Intelligent Systems for Molecular Biology & European Conference on Computational Biology (ISMB/ECCB), Prague, Czech Republic
"Kernelized rank learning for personalized drug recommendation"

## Teaching

2022 **Guest Lecturer**
University of Applied Sciences Technikum Vienna, Austria
- Special Areas of Applied AI (master's degree course)
  - Currently developping two lectures on deep learning in biotechnology and biomedicine that I will deliver in March 2022

2021 **Guest Lecturer**
Medical University of Vienna, Austria
- Biomedical Informatics and Genomic Medicine (master's degree course)
  - Developed and delivered the lecture and practical session on machine learning

2020–2021 **Guest Lecturer**
CeMM Research Center for Molecular Medicine, Austrian Academy of Sciences, Austria
- Machine-learning workshop for first-year PhD students
  - Developed and delivered the lecture and practical session

2016 **Teaching Assistant**
Department of Biosystems Science and Engineering, ETH Zurich, Switzerland
- Data Mining I (master's degree course)
  - Co-developed and delivered the practical sessions, marked homework and exam

2012–2013 **Teaching Assistant**

School of Information and Communication Technology, Griffith University, Australia

- Introduction to Programming (bachelor's degree course)
- Object-Oriented Programming (bachelor's degree course)
- Database Systems and Administration (bachelor's degree course)
  - Received a mean score of 4.78 (out of 5) and 5 award nominations: "Lukas was also an excellent tutor. He explains things so clearly and breaks up complex subjects into something simple which helped me understand topics better."
- Database Technology (master's degree course)
  - Received a mean score of 4.48 (out of 5) and 2 award nominations: "The nature of this lecturer was very engaging, and kept the students focused and interested."

## Student Supervision

2021–now Varun Sharma, PhD candidate (CeMM, Austrian Academy of Sciences, Vienna, Austria)
Project: Deep learning for single-cell transcriptomics

2020–now Stephan Reichl, PhD candidate (CeMM, Austrian Academy of Sciences, Vienna, Austria)
Project: Statistical and bioinformatics analysis of high-dimensional molecular data

2017 Elli Tzini, Master's degree student (ETH Zurich, Basel, Switzerland)
Project: Machine learning for personalised drug recommendation systems
*Elli continued her career as a machine-learning engineer.*

2016–2017 Udo Gieraths, PhD candidate (ETH Zurich, Basel, Switzerland)
Project: Machine learning for prediction of disease-linked genes
*Udo continued his career as a data scientist.*

2016 Caroline Weis, Master's degree student (ETH Zurich, Basel, Switzerland)
Project: Machine learning for prediction of disease-causing genomic variants
*Caroline continued her research as a PhD candidate at ETH Zurich.*

2013–2015 Mark Livingstone, Honours and PhD candidate (Griffith University, Southport, Australia)
Project: Machine learning for prediction of disease-causing genomic variants

## Institutional Service

2020–now Data ambassador and manager, part of a working group for implementing institute-wide "FAIR" data principles and transferring 650 TB of molecular biology data to new storage

2017 Contributed to the conceptualisation and presentation of a video game to popularise machine learning for medicine at the Swiss national scientific outreach event Scientifica

2012–2013 Founding chair of the IIIS/ICT Higher-Degree-Research Student Community

## Reviewing

Bioinformatics (rank 3 out of 59 mathematical and computational biology journals), Journal of Machine Learning Research, Journal of Molecular Biology, Scientific Reports, European Conference on Machine Learning

## Skills

Languages English – fluent, German – beginner, Slovak – native

Programming Python (incl. Scikit-Learn, PyTorch, Pandas, Statsmodels, NumPy) – advanced, Unix-based systems and bash – advanced, Java – advanced, R – intermediate, C and C++ – intermediate